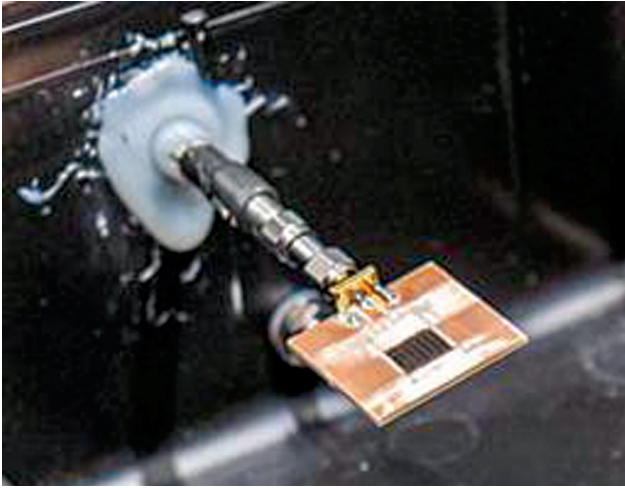


Artificial nose identifies scents accurately

Researchers at Norway's NTNU have developed a 33mm x 33mm microstrip antenna, partially coated with graphene oxide and the ionic polymer Nafion, which senses chemical



Artificial nose identifies scents accurately (Credit: <https://www.electronicweekly.com>)

compounds by analysing reflected signals over a 1GHz–8GHz frequency range. This antenna's general-purpose coating combines graphene and Nafion in a solvent that evaporates, forming a durable layer. During testing, it identified four alcohols with 96.7% accuracy and distinguished freshness in fruits. It can transmit data at 5.8GHz, showcasing dual use for communication and sensing. Researchers envision applications in environmental monitoring and medical diagnostics like scent detection by trained dogs.

Quantum driven transistors for ultra-efficient electronics

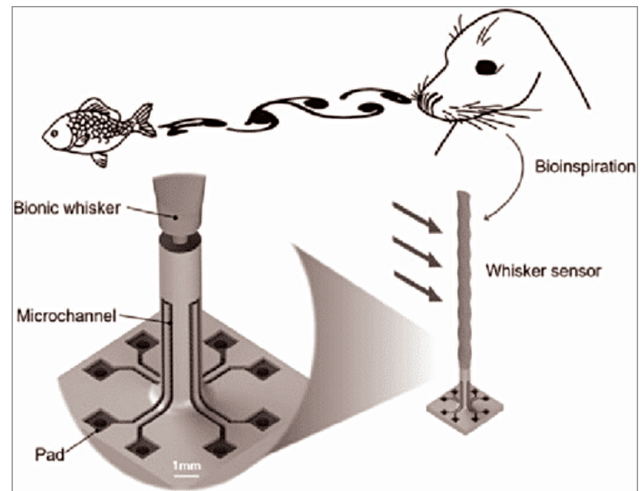
MIT researchers have developed advanced 3D transistors that overcome silicon's energy limits, using ultrathin materials and nanowire structures only a few nanometres in diameter. Conventional silicon transistors face 'Boltzmann tyranny,' a physical limit that prevents efficient low voltage switching. To bypass this, the transistors leverage quantum tunnelling, allowing electrons to cross energy barriers at reduced voltages, enhancing efficiency. These transistors are made from gallium antimonide and indium arsenide, which achieve sharp, energy-efficient switching. Fabricated with precise nanotechnology, they outperform traditional devices by 20 times. This innovation, funded partly by Intel, sets the stage for faster, more energy-efficient electronics beyond silicon's constraints.

Models help self-driving cars drive like humans

Researchers at TU Delft have developed a model that better captures human behaviour during motorway merging, addressing limitations in existing models. Unlike conventional approaches that assume drivers are always optimising for speed and safety, this model—created by postdoctoral researcher Olger Siebinga—emphasises risk perception and interaction. It accounts for nuanced behaviours like safety distance and right-of-way decisions, making it highly applicable for autonomous vehicle systems. Originally tested in a controlled, screen-based merging simulation, this model provides valuable insights for improving trust and acceptance of self-driving cars, with future plans to incorporate steering dynamics for even greater accuracy.

Seal-inspired sensors to safeguard offshore wind farms

Seals use their whiskers' unique twisted cylindrical shape to detect subtle vibrations, aiding in hunting within dark ocean depths. The team from UT Dallas has adapted



Schematic structure of the seal whisker-inspired flow sensors (Credit: [researchgate](https://www.researchgate.net))

this design into advanced sensors that detect water flow changes. Tested in simulated ocean conditions, these sensors offer crucial data to offshore wind operators, helping anticipate and prevent hazardous situations. Unlike sonar, which emits sound waves that disturb marine life, these vibration-sensitive sensors provide an eco-friendly solution for monitoring underwater turbulence, crucial for protecting floating wind turbines. This innovation supports offshore wind energy, where resilient, non-invasive monitoring is essential.